

Johansen Cointegration Test

Date: 07/09/26 Time: 11:52
Sample: 1/04/2010 12/30/2025
Included observations: 4159
Lags interval (in first differences): 1 to 4
Endogenous variables: EURUSD CHFUSD
Deterministic assumptions: Case 3 (Johansen-Hendry-Juselius): Cointegrating relationship includes a constant. Short-run dynamics include a constant.

Unrestricted Cointegration Rank Test (Trace)

Hypothesized No. of CE(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.** Critical Value
None	0.001874	14.07000	15.49471	0.0810
At most 1 *	0.001510	6.277750	3.841465	0.0122

Trace test indicates no cointegration at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Unrestricted Cointegration Rank Test (Max-eigenvalue)

Hypothesized No. of CE(s)	Eigenvalue	Max-Eigen Statistic	0.05 Critical Value	Prob.** Critical Value
None	0.001874	7.792254	14.26460	0.4002
At most 1 *	0.001510	6.277750	3.841465	0.0122

Max-eigenvalue test indicates no cointegration at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Unrestricted Cointegrating Coefficients (norma

EURUSD	CHFUSD
1.703259	14.09784
8.869297	-4.691356

Unrestricted Adjustment Coefficients (alpha):

D(EURUSD)	-0.000138	-0.000218
D(CHFUSD)	-0.000292	-3.46E-05

1 Cointegrating Equation
Log-Likelihood: 30845.75

Normalized cointegrating coeffici

EURUSD	CHFUSD
1.000000	8.276980
	(3.10027)

Adjustment coefficients (standard

D(EURUSD)	-0.000235	(0.00017)
D(CHFUSD)	-0.000497	(0.00018)